Guideline for Human Raters Proposition Based Theory Specification (PBTS)

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Introduction

Scientific theories aim to describe systematic regularities in the world. Often they concern causal relations between constructs. That is, many theories aim to capture e.g. how construct A influences construct B, construct B influences constructs C AND D etc. Scientific theories are according to classic approaches defined as a set of general implications (if - then statements) that describe the relation between propositions (i.e. statements that could be true or false). In reality, however, most often theories are presented as narrative descriptions. This means that the general implications and propositions of a theory are not fleshed out explicitly. The proposition based theory specification method (PBTS) allows to explicate the implications and propositions contained in a theory in a formal and structured way.

Definitions
We refer to *propositions* as statements that can be true or false.
We refer to *implications* as statements that relate propositions in the form of IF- THEN statements. Implications are also statements that can be true or false. They therefore are a more complex form of propositions.
Propositions contain statements concerning concepts. Concepts can be not directly measurable (latent) variables (often referred to as constructs, e.g., intelligence, feeling of frustration) and directly measurable variables (such as behavior or properties of the person or situation etc., e.g., aggressive behavior, height of the person, temperature in the room). Concepts can therefore refer to constructs, behaviors, situations, and also other types of variables.

IF proposition A [*antecedence*] THEN proposition B [*consequence*]

An example from psychology is:
IF (a person feels frustrated[=true]) THEN (he/she shows aggressive behavior[=true])

This means that for all possible situations and all individuals, if frustration is true then aggression is also true, or stated differently, frustration always leads to aggression. The statement IF (proposition A) is a short form for writing IF (proposition A=true), the part in squared brackets above [=true] can be omitted in such cases.

In psychological theories, propositions typically contain states of unobservable concepts (i.e. latent constructs), sometimes also observable concepts. For latent constructs, operationalizations have to be provided to be able to decide whether a statement is true or

false. Also for observable concepts, it has to be defined when a proposition is considered true or false.

To allow for testable predictions, a definition of a theory should include

a) all relevant propositions and the implications that describe their relations,
b) definitions of all measurable and not directly measurable concepts that are included in the propositions and
c) (examples for) valid operationalizations / measurement methods of these concepts including criteria for deciding when the proposition is considered to be true or false. Sometimes it is helpful to provide operationalizations on a population level (e.g. more than half of the subjects chose option A over option B) to allow for errors and statistical testing.

Typically it is aimed that definitions and operationalizations are in line with common definitions and operationalizations used in the literature and validated before.

Note that often issues of measurement and operationalization of concepts are not considered part of a theory. PBTS explicitly makes their specification part of the specification of a theory. This should foster testability of theory, replicability, and efficient theory development by solving the underdetermination problem identified in the literature (see, for example, Duhem-Quine thesis, Meehl, Lakatos).
For example, to make the implication introduced above testable, we (a) need to define how to measure the construct “a person feels frustrated” (and potentially define a threshold value from which we consider this to be true, in case a continuous measure is used). Furthermore, we (b) have to specify how to measure the observable variable aggressive behavior and how to decide whether a behavior is considered aggressive or not. Spelling out the example already leads to interesting insights. For example, we learn that the current specification of the “theory” implies binary states of frustration (yes vs. no) and no gradual increase in the likelihood of aggressive behavior with increasing frustration. Furthermore, it becomes clear that for a theory to make clear-cut predictions, we have to invest considerable thought in operationalizations and measurement of the relevant concepts.

In order to capture complex and computational theories as well as relations between continuous variables/concepts, we allow the antecedence and consequence to also include multiple propositions that can be combined by any logical operators (AND, OR, NOT), or qualified by probabilistic statements, or any functions. Introducing new operators is possible too, as long as they are defined in step (3), below.

For example:
If (construct A) *AND* (situation B) THEN (behavior C)
If (construct A) *AND NOT* (construct B) THEN (construct C) *OR* (construct D)
If (construct A) THEN (construct B) = (construct C) * (construct D)^(construct E)
If (construct A) THEN (*5% of the population shows* construct B)
If ((construct A) and (construct B) *are balanced*) THEN (construct C)
[comment: it has to be explicitly specified when the operator (e.g. “*are balanced”*) produces true or false values in step 3, below)]

Each proposition should be put in parentheses to improve clarity. Operators should be written in italics. Each concept / proposition and non-standard operator has to be listed and defined in step 3 (below).

Moreover, theories can include further auxiliary assumptions that are needed to make a theory testable or applicable. These can include pre-conditions under which the theory is applicable or can be tested that are not an inherent part of the theoretical assumptions (i.e. propositions) and operationalizations/definitions.

Theories in psychology are often formulated verbally as narratives and not in the form of explicit propositions or equations as described above. They often contain multiple complex relations between constructs.

The aim of this theory specification methodology is:

- to transform the verbal specification of a theory into formal propositions of the form described above,
- to provide explicit definitions and (examples for) valid operationalizations of each concept / proposition,
- to find the simplest possible specification of the theory in the form described above that still contains all relevant relations of the theory,
- to find a specification that is so simple that any scientist can generate it and read it,
- to generate a specification that is objective in the sense that for any situation the prediction of the theory can be directly derived by anybody [or it is clear that the theory is not applicable or makes no predictions],
    - note that this explicitly does **not** mean that the theory is correct in these predictions
    - the aim is to learn from potential counter-examples / "falsifications" to improve the theory,
- to generate a specification that invites critical testing and can be easily revised and improved in an iterated process,
- to foster testing of theories, theory revision, theory integration, more efficient science and cumulative development of knowledge.

Formal Theory Specification

The suggested methodology describes a collective approach that should increase objectivity and allow to estimate reliability of the specification process. This will involve the steps described below. In some cases, steps might also be repeated and the results of previous steps might be modified to come to a good solution for all parts. It is advisable to conduct the theory specification process with multiple individuals and coordinate with each other. For later steps it is useful to complete them in dyads, therefore an even number of raters is recommended. Typically, two persons will suffice to conduct the procedure. Involving both a relative novice and an expert concerning the theory as team members can be helpful, as it offers the advantage of combining different perspectives - such as an approach that is closer to the writing and an approach that includes further knowledge.

Helpful guidance:

For the current project, we recommend that the theory under consideration should first be specified in its original or early form, as it was initially proposed by the author of the theory. Later findings, revisions, and critical insights should not be addressed in depth yet—the focus should remain on the original formulation of the theory itself. It is possible to specify extensions and modifications of the theory in a second application of (parts of) PBTS using essentially the same methodology, which is often helpful to clearly describe the development of the theory. This first specification aims to represent the core of the selected theory as accurately as possible as it is usually understood and tested in the literature. In case the original formulation of the theory was very long (e.g. provided in a book), the specification does not necessarily need to strive for exhaustive detail. It is intended as a starting point that researchers can expand upon in further steps, addressing potential issues and delving into greater depth where needed.

General steps proposed by PBTS

Preparation

(Step 0) Select the theory and the version of the theory to be specified.
(Step 1) Select relevant sources for the theory specification and agree on these sources with fellow raters.

Independent Theory Specification

(Step 2) State a finite set of implications (i.e., IF- THEN statements) and all included propositions that together fully describe the theory.
(Step 3) Define and operationalize all concepts / propositions as well as all non-standard / new operators (if used) that are listed under Step 2.
(Step 4) Specify auxiliary assumptions and antecedence conditions of the theory.
(Step 5) Double-check for problems and subjectively imposed aspects (e.g. aspects that were not specified by the author of the theory) in the theory specification and note important insights.

Generate Consensus Theory Specification & Rating of the Theory

(Step 6) Rate various aspects of the theory.
(Step 7) Evaluate rater agreement and achieve consensus on a final version.
(Step 8) Analyze theory specification issues and classify implications.

Typically, the theory specification should be done in teams of at least two people. It is crucial that the selection of the Theory and Version as well as the source selection (Steps 0 and 1) are conducted jointly by all raters. To be able to measure reliability of the assessment, Steps 2 to 5 are first conducted individually. Then theories are individually rated based on these experiences, individual specifications are compared to generate reliability measures and they are integrated into a joint final solution in Steps 6 to 8.
In case of teams that consist of more than 2 persons, for Step 7, it is advisable for raters to work in dyads first and come together for consensus building with all team members only in a further step to be able to generate agreement / reliability measures for each dyad.

The results of Steps 0 to 8 should be recorded in the theory specification template (containing multiple tables) provided in a Google Drive folder under the link: https://drive.google.com/drive/u/0/folders/1IYQ1pdMqWrfXYtUWCzUrQgkyAfBIiK-p . The

folder including the templates should be downloaded when using them for theory specification. The tables for documenting steps 0 to 5 are listed in the following since they also include examples which help to understand the process. Further, more detailed instructions are provided below the tables. Please read this guideline first before starting theory specification.

Theory Specification Template

Table 1: THEORY AND SOURCE SELECTION (specified in Steps 0 and 1) =version of the theory and text sections used for the following theory specification Please state the theory and version of the theory you want to specify. Cite the sources used, indicating the pages and line details of the relevant sections for the following theory specification or link to the shared document in which the text sections are contained. The raters should have agreed on the sources before conducting further steps. However, sometimes necessary updates to the list might become clear only after conducting further steps.

Theory and Version of the Theory to be specified:		
Nr	Source	Pages and Lines used
1		
2		
3		
Alternatively - Link to shared document:		
Explanation for Source Selection (if necessary):		

Table 2: PROPOSITIONS / IMPLICATION (Specified in Step 2)
= Main statements that describe the theory.
Please formalize the propositions / implications using IF ... THEN ... statements, where IF = antecedence, THEN = consequence. Use also OR, AND or any logical or mathematical operators if you need it. In case you use uncommon operators, please also explain the operators briefly (in Table 3). If an implication can be directly retrieved from the formulation of the theory and is included in the sources that were determined in Step 1, provide a reference. If the proposition cannot be directly extracted or you have to impose further assumptions to do so, add a marker “DER” (derived) in the comments section and explain briefly.

Example "Facial Feedback Hypothesis":
(Antecedence:) IF (the facial muscles related to emotional expressions *are activated*) AND NOT (monitored)
(Consequence:) THEN (positive emotions *are induced*) AND (stimuli *are evaluated more positively*).

Nr	IF	Then	Reference / Comments / Markers
1			
2			

Table 3: DEFINITIONS AND OPERATIONALIZATIONS OF CONCEPTS / PROPOSITIONS / OPERATORS (specified in Step 3)
= further specification of concepts / propositions / operators used within the formulation of the implications in Step 2.
Please, type in a concept or proposition you have used in your implications in Step 2, its verbal definition and operationalization. Think about what you would need to test the theory's proposition to be able to determine whether they are true or false.
Add a reference if you could directly extract the definition and the operationalization from the formulation of the theory, hence being included in the sources that were determined in Step 1. If the definition/operationalization cannot be directly extracted from these sources or you have to impose further assumptions to do so, add one of the following markers in the comments section and explain briefly.
"DER” (Derived): Indicates that the concept is not present in the determined sources and additional sources or personal knowledge were used to derive information; provide the source or a brief explanation.
"MIS” (Missing): Indicates that the concept is missing in the determined sources and could not be found quickly in additional sources. Moreover, it could not be supplemented through prior knowledge.

Example "Facial Feedback Hypothesis"
Note, that 7 concepts and operators are contained in the example, only the first three are specified in the following.

Nr.	concept label / proposition / uncommon operator	Verbal definition	Measurement/ Operationalization
1	facial muscles related to emotional expressions	Facial muscles related to emotional expressions	zygomaticus major tension measured by Electromyography method (EMG)
2	the facial muscles related to emotional expressions *are activated*	Activation of facial muscles expressions	Is the EMG firing rate of facial muscle expression significantly higher in Condition A than in Condition B AND effects size d >= .30
3	monitored	A person is aware that (s)he is watched by others and monitors consciously his/her facial expressions to show the emotions (s)he likes to show	Self-report: Were you aware that you are filmed or watched by others? YES / NO

Nr.	concept label / proposition / uncommon operator	Verbal Definition	Measurement/ Operationalization	Reference / Comments / Markers
1				
2				

Table 4: AUXILIARY ASSUMPTIONS (specified in Step 4)
[only if necessary - potentially skip]

= assumptions that are secondary to the theory under investigation. They describe necessary pre-conditions to test a theory or boundary conditions. Only aspects that are not already covered in step 3 should be listed here. All auxiliary assumptions mentioned by the authors of the theory should be covered. Further auxiliary assumptions and pre-conditions can be listed, if they can be derived from the literature or on theoretical / logical grounds. A brief definition and a reference should be provided for the auxiliary assumptions made by the author of the theory. Further auxiliary assumptions that have been derived should be marked with “[DER]”, defined and explained. Universal auxiliary assumptions that apply to almost every theory should be separated from theory-specific auxiliary assumptions by adding “UNIV” or “SPEC” to the “[DER]” marker, respectively. The reasoning for both of them should be briefly explained.

Example "Facial Feedback Hypothesis":

Auxiliary assumption(s): Implication 1 only holds, if the muscles were not activated before by another cause or Implication 1 only holds in Western countries.

Nr.	Description	Definition (Optional: Operationalization)	Reference / Comments / Markers
1			
2			

Table 5a: Problems (specified in Step 5)

= Basic requirements for any scientific theory to hold are the absence of tautologies and logical contradictions. In addition, precise definitions of all aspects of the theory should be provided. Sometimes problems only become apparent if a theory is formally specified according to the steps above.

Are there any tautologies in the specification of the theory or is there a risk that a tautology might appear for some operationalizations of variables? Note each potential problem in a separate line and mark them with [TAU].

Are there any contradictions in the specification of the theory or is there a risk that a contradiction might appear for some specifications? Note each potential problem in a separate line and mark them with [CON].

Are there issues in the verbal description of the theory that are underspecified to a degree that deriving implications and propositions is hard to impossible? Issues that are already solved by operationalizations and definitions are excluded here. List them, one per line, below and mark them with [USPEC]. Provide a short explanation in a separate column.

Nr.	Description	Reference / Comments / Markers
1		
2		

Table 5b: Insights, Further directions and Open Questions (Specified in Step 5):

In the process of theory specification, one often gains insights concerning

(i) particularly important and unexpected aspects of the theory,
(ii) aspects that would be important to specify further, and
(iii) direction for further theory development or theory integration.

These insights should be listed below using one line each.

Nr.	Description	Reference / Comments / Markers
1		
2		

Detailed Explanations and Instructions for Steps 0 to 8

(Step 0) Select the theory and the version of the theory to be specified.

Before starting the theory specification, the raters should agree on the following questions: Which theory should be specified? Specify the name of the theory, the author and the original reference.

If applicable: which version of this theory (e.g., original version, revised version, latest version) should be specified?

Both answers should be documented in Table 1. The answers, however, might also be revised after a more extensive literature search in step (1).

(Step 1) Select relevant sources for the theory specification and agree on these sources with fellow raters.

Further instructions and explanations:

(a) Source Selection
   (i) for the specification of a theory limit the sources to 2 or 3, for example, original article, overview textbook chapters (e.g., for this project, Handbook of Theories of Social Psychology: Volume 1 & 2 by Van Lange et al. (2013), https://doi.org/10.4135/9781446249215), journal articles, or recent works that present the theory to be examined. *For the theory specification challenge please always use the handbook chapter as one source. Typically, one would add the original article as a second source and give it particular priority in the specification process.*
   (ii) If the original theory was extremely comprehensive and for example formulated extensively as a book: For pragmatic reasons, the specification could focus on the core of theory as it is reflected in typical journal articles and summary book chapters. Extending this core specification by further aspects that might additionally be contained in an extensive book could be left to further research. However, such decisions (i.e. against using the original book or article) need to be explicitly documented in Table 1.

(b) Detecting Relevant Text Snippets
   (i) Which part of the text contains descriptions of the causal relations that are proposed by the theory (i.e. their implications and propositions)? This includes statements concerning all (relevant measurable and non-measurable) concepts and their relations, including all mentioned qualifiers.
   (ii) Which part of the text contains descriptions and definitions of the concepts that are included in these propositions?
   (iii) Which part of the text contains descriptions of operationalizations of the concepts included in the theory or allows deriving or retrieving them?
   (iv) Which part of the text contains further auxiliary assumptions that are important for applying or testing the theory?

(v) Please provide the exact position of the relevant sections (e.g. from page 1, line 1 to page 2 line 20).
(vi) The text snippets for answering the questions above can (and will in most cases) be overlapping.
(vii) The text snippets should always include full sentences, often they will concern whole sections.
(viii) It is important that all causal relations, operationalizations, definitions, and auxiliary assumptions are covered and understandable. Therefore, it is preferred that the text snippets are not too narrow and can also include some explanatory text around them.

(c) Agreement on Source Selection: Before specifying the theory, it is important that the raters agree on the sources being used, as the specification depends on the selected texts and this will reduce the time spent on the following steps.
(i) Ideally, use a shared version of Table 1 for documenting sources. All raters can input what they consider relevant sources and parts.
(ii) Reach an initial agreement on which text sections to be used for the specification based on discussing the issue. In case of doubt, include more text.
(iii) Document the result of this agreement in the final version of Table 1. You can either name the sources including page and line information or insert the link to the shared document.

(Step 2) State a finite set of implications (i.e., IF- THEN statements) and all included propositions that together fully describe the theory.

Further instructions and explanations:

(a) Propositions contain statements concerning measurable and non-measurable concepts that can be true or false.
(b) Propositions can be combined into more complex propositions that then represent the antecedence and the consequence, respectively.
(c) Propositions can be combined by any logical or mathematical operators, qualifiers, or if necessary new/uncommon operators (that have to be defined and operationalized).
(d) Note, that all propositions must be fulfilled for the theory to hold. Therefore all propositions are implicitly linked by a logical AND, this allows for the following simplifications:
(i) for propositions that include the same “if” statement, write them in one proposition using AND in the “then” statement to connect the consequences (if A then B / if A then C → if A then B AND C)
(ii) for propositions that include the same “then” statement, write them in a proposition using OR in the “if” statement connecting the antecedents (if A then C / if B then C → if A OR B then C).
(e) It is often possible to make certain conditions/ statements of a theory either part of the propositions or the operationalizations. To keep the propositions simple, these conditions should be part of the operationalizations if they do not belong to the core causal relations of the theory.

(f) Specific predictions concerning functional relations between concepts can be put in the “then” part of the implication. For example: “if” (A takes values in the Range [0-1]) “then” (B increases with increasing A)). A similar logic can be applied for more complex functional forms by changing the “then” component as follows: “then” (A = f(B)); with f() indicating a weakly monotonic / linear / exponentially increasing function (as specified in the Table for uncommon functions in step 3); or “then” (A = B^alpha); with alpha > 0.
(g) Functions can have different degrees of precision and it is fine to start with relatively vague functions (e.g., f() … weakly monotonic increasing function vs. linearly increasing function).

Refresher on logical operators:

- AND (Conjunction): True only if all connected propositions are true
  - IF A **AND** B THEN C → Both A and B must be true for C to occur
- OR (Disjunction): True if at least one connected proposition is true
  - IF A **OR** B THEN C → Either A or B or Both must be true for C to occur
- NOT (Negation): True when the connected proposition is false, and vice versa
- IF A AND NOT B THEN C →A must be true and B must be false for C to occur

Pragmatic recommendations

- identify the core propositions and implications of the theory and put them in a numbered list of IF - THEN statements. Use a separate line for each IF - THEN statement.
- Use parentheses for a proposition and italicize uncommon operators.
- In each line add the text snippet including the reference where you got each of the implications from in a separate column.

(Step 3) Define and operationalize all concepts / propositions as well as all non-standard / new operators (if used) that are listed under Step 2.

Further instructions and explanations:

(a) Extract all concepts / propositions and non-standard operators from the list of implications (Table 2) and list one per line in Table 3.
(b) Standard operators do not need to be defined and operationalized.
  (i) Common operators are links or functions that occur as standard in logic or mathematics (*AND, OR, NOT, +, -, , /, ^)
(c) Add definitions for each concept / proposition as provided (or referred to) by the author of the theory.
(d) Add operationalizations for each proposition as provided (or referred to) by the author of the theory.
(e) Add direct references/ text snippets for (a), (c), and (d) as references from the sources determined in step 1.
(f) Optional: Theory Advancement (*it is recommended to also implement this step for the theory specification challenge to generate a full formal specification of the theory*)
  (i) In case, definitions are not provided by the author, derive them from your previous knowledge or the scientific literature in psychology. Mark them with “[DER]” and provide the source or a brief explanation in an extra column if possible.

(ii) In case, operationalizations are not provided by the author, derive them from your prior knowledge or the scientific literature in psychology. Preferably use operationalizations that were used by the author of the theory in other empirical work or other authors that claimed to test the theory empirically. Mark them with “[DER]” and provide the source in an extra column or a brief explanation.

(1) Double-check whether these operationalizations have generated / can be expected to generate the typically found effects in the standard paradigms.

(2) It is preferable to have various operationalizations that can be applied in different situations since otherwise the theory often cannot be applied and the empirical content of the theory is reduced. If possible provide operationalizations for

a) experimental situations (manipulation & manipulation check)
b) surveys (e.g. questions, this could also be the same as before)
c) practical applications / behavior outside the lab (e.g., external ratings of the situation or the person that can be applied post-hoc or using observation).

Pragmatic recommendations:
This step can easily take overly much time. We recommend to set an explicit time limit for this step (e.g., max two hours overall). For all definitions and operationalizations of propositions that cannot be derived directly from the determined sources a “MIS” (missing) can be added.

(Step 4) Specify auxiliary assumptions and antecedence conditions of the theory.

Often measurement / operationalization issues addressed under Step 3 are considered auxiliary assumptions. These do not have to be listed here again. In this section, only further assumptions and pre-conditions of the theory to hold can be listed.

(a) Auxiliary assumptions and pre-conditions that are mentioned by the author of the theory should be listed in a numbered list, one per line. They concern requirements that should be met for all propositions / implications. For each line, an explanation and/or text snipped including reference should be added in a separate column that describes the origin of the auxiliary assumption. Moreover, a brief definition of that assumption should be given.
An example of an auxilliary assumption could be: “theory is only applicable in Western countries”, with “Western countries” defined in the column next to it.

(b) Optional: Theory Advancement
Further auxiliary assumptions and pre-conditions can be listed, if they can be derived from the literature or on theoretical / logical grounds. Auxiliary assumptions should be used sparsely but it is good to explicate them, if it can be assumed that most researchers would implicitly share them. They should be marked with “[DER]”, and a reference and / or explanation should be provided.
Universal auxiliary assumptions that apply to almost every theory should be separated from theory-specific auxiliary assumptions by adding “UNIV” (for

universal assumptions) or “SPEC” (for theory-specific assumptions) to the “[DER]” marker. Examples could be - the theory only holds if:

(individual was attentive) AND (individual understood the structure of the task).

For propositions listed here, operationalizations should be provided as well if possible (e.g., in experiments: participant passed attention check and comprehension questions).

(Step 5) Double-check for problems and subjectively imposed aspects (e.g. aspects that were not specified by the author of the theory) in the theory specification and note important insights.

Ideally, all relevant information for a theory is provided by the theories’ author. This makes it possible to identify the propositions / implications and their relations as well as operationalizations and definitions for all variables involved.
In the process of formalizing theories, issues that remained underspecified and logical errors in the original verbal specification can sometimes be detected. Detecting these problems provides important directions for advancing the further theory specification (see optional parts). Please report problems in Table 5a and important insights, further directions and open questions in Table 5b.

As already introduced above, basic requirements for any scientific theory to hold are the absence of tautologies, logical contradictions, and a sufficiently precise definition of all aspects of the theory. Sometimes, problems only become apparent when a theory is formally specified following the steps above.

A tautology means that the antecedence (IF propositions) and the consequence (THEN propositions) of an implication contain similar, overlapping, or even the same variables. Thus, different names of variables might essentially refer to the same latent construct or different constructs might be operationalized in very similar manners (i.e. measured with very similar questions). Thereby, a risk of tautologies often only becomes visible when examining the details of operationalizations of variables.

A contradiction means that logical contradictions can be derived from a set of theoretical implications. In many cases, contradictions concern predictions that logically exclude each other (e.g., the implications (1) IF (feeling of frustration) then (aggression) and (2) IF (feeling of frustration) then (NOT aggression) contradict each other).

An underspecification of implications and propositions can concern various aspects that are left open regarding the specific relation of the constructs. Some of these issues are solved by specifying operationalizations. The remaining issues that are critical for deriving clear-cut propositions and implications should be listed here. Further suggestions for improvement that would make the theory more precise but are not necessarily required for specifying implications can be added as described below (i.e. theory advancement).

(a) Are there any tautologies in the specification of the theory or is there a risk of tautology with some operationalizations? Note each potential problem in Table 5a in a separate line and mark them with [TAU].
    (i) Optional: Theory Advancement
        (1) Try to find definitions and operationalizations that are as distant as possible from each other to avoid tautologies. Add them as alternative possibilities to the respective table for example in the comments section (making them with DER as derived).

(b) Are there any contradictions in the specification of the theory or is there a risk of contradiction with some specifications? Note each potential problem in Table 5a in a separate line and mark them with [CON].
    (i) Optional: Theory Advancement
        (1) Try to find more precise specifications that avoid the risk of contradictions or drop some of the propositions. Add these solutions as comments or alternative possibilities to the respective tables (making them with DER as derived).

(c) Are there issues in the verbal description of the theory that are underspecified to a degree that deriving implications and propositions is hard to impossible? Issues that are already solved by operationalizations and definitions are excluded here. List them, one per line, in this section and mark them with [USPEC]. Provide a short explanation in a separate column. Without aiming to be exhaustive, here is a list of issues that have been detected in past attempts.
    (i) Conceptual clarity: Sometimes in a verbal theory specification it is hard to differentiate constructs from each other.
    (ii) Timing issues: Often psychological processes are dynamic and it might not be easy to derive specific predictions concerning the exact timing or development of variables over time. A theory does not have to provide detailed predictions, as long as it makes clear and unambiguous predictions (e.g., B happens after A / as a result of A is fine).

(d) To fully specify a theory, often some aspects have to be imposed since they are left open by the author of the theory. It is fine to decide between the different possibilities to fill these loopholes and to further specify the theory. However, it is important to explicitly mark/note that such decisions were made. Also, reasons should be provided for making the choice in one way or the other.
    (i) Please double-check for all propositions, operationalizations, definitions, and auxiliary assumptions, whether you have filled loopholes by imposing aspects in theory specification steps 2 to 4, and mark all of them, if not already done, with “[DER]”.

(e) Optional: Theory Advancement - Insights, Further Directions, and Open Questions
In the process of theory specification, one often gains insights concerning
    (i) particularly important and unexpected aspects of the theory,
    (ii) aspects that would be important to specify further, and
    (iii) direction for further theory development or theory integration.
These should be listed in Table 5b using one line each.

Continue with Step (5) only, if you are satisfied with your overall theory specification and feel that you have finished the independent part of the theory specification procedure.

(Step 6) Rate various aspects of the theory.

Table 6: Theory Ratings

Please provide overall ratings concerning the following aspects of the theory you specified in steps 2 to 5. These ratings should be based on the version of the theory and sources specified in steps 0 and 1 and reflect your experiences from individual theory specification in steps 2 to 5. Aspects that you added to the theory or derived from other sources should not be considered in the ratings. Please indicate whether you agree or not to the following statements using the following 7-point scale: 1: Strongly Disagree 2: Disagree 3: Somewhat Disagree 4: Neither Agree nor Disagree / Neutral 5: Somewhat Agree 6: Agree 7: Strongly Agree	
<u>6.1 Theory Specification</u>	
1) The core mechanisms that the theory proposed are well described by the author.	
2) The propositions and implications that constitute the theory are clearly stated.	
3) The relations between all concepts are articulated in an unambiguous manner.	
4) Definitions for all relevant concepts are provided.	
5) Operationalizations for all relevant concepts / propositions are provided.	
6) The various aspects of the theory are overall very well specified by the author.	
7) The original formulation of the theory contains risks for tautologies to appear.	
8) The original formulation of the theory contains risks for inconsistencies.	
9) The original formulation of the theory contains risks for underspecification.	

Please provide overall ratings concerning the following aspects of the theory you specified in steps 2 to 5. These ratings should be based on the version of the theory and sources specified in steps 0 and 1 and reflect your experiences from individual theory specification in steps 2 to 5. Aspects that you added to the theory or derived from other sources should not be considered in the ratings.

Please indicate whether you agree or not to the following statements using the following 7-point scale:

1: Strongly Disagree
2: Disagree
3: Somewhat Disagree
4: Neither Agree nor Disagree / Neutral
5: Somewhat Agree
6: Agree
7: Strongly Agree

6.2 Subjective Assessment of Theory Specification	
1) It was easy for me to extract the propositions and implications from the authors' descriptions of the theory.	
2) Extracting definitions for the relevant variables was easily possible.	
3) Extracting operationalizations for the relevant variables was easy.	
6.3 Properties of the resulting specified versions of the theories	
6.3.1 Complexity	
1) The theory contains many variables and relations compared to other theories in social psychology.	
2) The theory contains many complex relations between variables compared to other theories in social psychology.	
3) The theory is overall very complex compared to other theories in social psychology.	
4) How many implications are included in the theory? (lines in Table 2 for implications)	_____
5) How many concepts / propositions / idiosyncratic operators are contained in the specification? (lines in the Table 3 for concepts / propositions)	_____

Please provide overall ratings concerning the following aspects of the theory you specified in steps 2 to 5. These ratings should be based on the version of the theory and sources specified in steps 0 and 1 and reflect your experiences from individual theory specification in steps 2 to 5. Aspects that you added to the theory or derived from other sources should not be considered in the ratings.

Please indicate whether you agree or not to the following statements using the following 7-point scale:

1: Strongly Disagree
2: Disagree
3: Somewhat Disagree
4: Neither Agree nor Disagree / Neutral
5: Somewhat Agree
6: Agree
7: Strongly Agree

6.3.2 Empirical content of the theory
generality (i.e. inclusiveness of the if-components / antecedent) and precision (i.e. the exclusiveness of the then-components / consequence).

6.3.2.1 Generality

1) The theory is very general in that it can be applied to many different situations and persons.	
2) There are boundary conditions explicated that reduce the generality of the theory.	
3) The antecedence of the theory (IF component) is broad and does not constrain the application area substantially.	
4) The auxiliary assumptions constrain the application area of the theory substantially.	
5) The operationalizations provided allow to test the theory broadly and in many situations.	

6.3.2.2 Precision of the theory

1) The theory is very precise in its predictions.	
2) The THEN component of the theory specifies clearly what should happen.	
3) The THEN component of the theory is restrictive in that it forbids many alternative consequences.	
4) The theory makes testable predictions on multiple dependent variables.	

Please provide overall ratings concerning the following aspects of the theory you specified in steps 2 to 5. These ratings should be based on the version of the theory and sources specified in steps 0 and 1 and reflect your experiences from individual theory specification in steps 2 to 5. Aspects that you added to the theory or derived from other sources should not be considered in the ratings.

Please indicate whether you agree or not to the following statements using the following 7-point scale:

1: Strongly Disagree
2: Disagree
3: Somewhat Disagree
4: Neither Agree nor Disagree / Neutral
5: Somewhat Agree
6: Agree
7: Strongly Agree

Uniqueness of theory predictions	
1) The theory makes many predictions that differ from predictions of other theories I know of.	

(Step 7) Evaluate rater agreement and achieve consensus on a final consensus version.

The purpose of this assessment is to systematically capture the content-based agreement of (7.1) Implications / Propositions and (7.2) Definitions/Operationalizations formulated by two or more raters, to evaluate any differences, and (if possible) to find a final consensus version (7.3) raters agree on.
It is recommended to have the raters work in dyads for the following steps. Once dyads have agreed on their final version, there is an opportunity to exchange with all raters about their final versions and, ideally, come to an overall consensus.

(7.1) Implications / Propositions

1) **Listing implications / propositions**
    a) Each implication / proposition created by Rater 1 (i.e. each line from his / her Table 2) is entered in a separate row of Table 7.1 below. Rater 1 enters their implications / propositions under “Version Rater 1”.
2) **Matching implications / propositions**
    a) Rater 1 lists the implications / propositions by Rater 2 under “Version Rater 2”. Rater 1 tries to match the implications / propositions by Rater 2 as well as possible to the rows of “Version Rater 1”. Implications / propositions that address the same aspect (might be more than one implication / proposition) should be in the same line even if they are not fully identical. It is also possible that a implication / proposition from Rater 2 can be matched with two implications / propositions from Rater 1, allowing for the proposition from Rater 2 to be listed twice. If Rater 2 addresses a new aspect that is not part of Rater 1’s implications / propositions, a new row should be created.
    b) If some aspect is only captured by one of the raters, the corresponding cell for the other rater remains empty.
    c) Number the implications / propositions as listed by each rater in Step 2 in their Table 2, respectively. Place the corresponding (line) number in parentheses behind the respective implication / proposition in each rater's version.
    d) Rater 2 double-checks the matching. In case of disagreement, Rater 2 makes an alternative suggestion and Rater 1 and 2 find a common solution for the matching.

3) **Assessment of the Agreement:**
    a) We aim to get two independent ratings of agreement of the implications / propositions based on the matched list in Table 7.1 generated in the previous matching step. Therefore, rater 1 and 2 should provide their agreement ratings in Table 7.1 independently. For this, it will be necessary to generate two copies of the table and to copy the ratings in a final version of the table only after the independent ratings have been completed.

b) Each rater assesses each row for content-based agreement. It is recommended that ratings are not too strict so that minor differences are not attributed to the author of the original theory, especially if they reflect natural variation in expression. Linguistic differences in wording, such as synonyms, capitalization, and the use of definite or indefinite articles (e.g., “a person” vs. “the person”), should be disregarded. Furthermore, consider the following cases as complete agreement (with an explanation in the comments):
    i) If one rater expresses a proposition as a single statement while the other splits it into two or more propositions.
    ii) If a statement appears in one rater's proposition but is embedded within the verbal definition/operationalization of the other.
c) The following rating scale should be used:

- **4: Full Agreement** - Propositions are entirely aligned in their content meaning, even if synonyms are used.
- **3: 75% Agreement** - Propositions are largely aligned in content, with only minor differences.
- **2: 50% Agreement** - Propositions show a moderate degree of alignment, with some significant differences.
- **1: 25% Agreement** - Propositions show little alignment in content, with several notable differences.
- **0: No Agreement** - Propositions show no content-based agreement.

d) Please document any considerations and uncertainties in the comment column.

4) Discussion of the Agreement

a) The two raters should combine their ratings in the final table and meet up to discuss their ratings and comments. Any further insights and comments could be added in the comments section. We will use the average rating as overall agreement measure and also calculate the agreement.

5) Statistical Evaluation

a) TBA (e.g., Compute the Intraclass Correlation Coefficient (ICC) and means for the ratings to assess the overall agreement)

Table 7.1 Evaluation of Rater Agreement on Implications / Propositions (from Tables 2)

Nr.	Proposition - Rater 1	Proposition - Rater 2	Rating Rater 1	Comment
1	*Example:* *IF (the facial muscles related to emotional expressions are activated) AND NOT (not monitored) THEN (positive emotions are induced) AND (stimuli are evaluated more positively*) (1)	*IF (the facial muscles are activated) AND NOT (not consciously monitored) THEN (positive feelings are induced) AND (stimuli receive a more positive evaluation*) (1)	2	R1: Synonyms: “emotions” and “feelings”; “are evaluated more positively” and “receive a more positive evaluation” Differences: *(the facial muscles **related to emotional expressions)*** in V1 vs. not in V2; (... **consciously** monitored) in V2 vs not in V1

Please rate the agreement between the propositions by entering the following number:

4 - Full Agreement
3 - 75% Agreement
2 - 50% Agreement
1 - 25% Agreement
0 - No Agreement

Nr.	Implication / Proposition - Version Rater 1	Implication / Proposition - Version Rater 2	Rating Rater 1	Rating Rater 2	Comment
1					
2					
3					

(7.2) Definitions/Operationalizations of Concepts / Propositions / Operators

1) The same procedure as described in (7.1) should be applied to the definitions and operationalizations of concepts/ propositions / uncommon operators including Listing, Sorting & Matching, Assessment and Discussion of the Agreement and Statistical evaluation.
    a) Please note that the concept label, verbal definition, and operationalization are to be put in the table as shown below.
    b) The numbering from Step 3 (i.e. line numbers on Table 3 of each rater) should also be placed in parentheses behind each corresponding element.
2) Assessment of Agreement - additional remarks
    a) Please note again that the evaluation in this step should not be overly strict.
        i) If Rater 1 has defined and operationalized a concept that Rater 2 has divided into two or more statements, these can be considered as matching.
        ii) Similarly, if Rater 1 has already addressed part of the concept in the propositions (Step 2), while Rater 2 only incorporates it in Step 3, this can also be considered a complete agreement.
    b) For cases where both raters either put in DER or MIS, rate as 4 (Complete Agreement). Both imply that the raters could not find definitions or operationalizations in the text.
    c) If a concept was used by only one rater, do not rate the agreement of its definition and operationalization, as it is already captured through the rating in Step 7.1. Rate as Not Applicable (=NA) instead.
    d) The following rating scale should be applied **for verbal definition and operationalization (not concept labels!)**.

- **4: Full Agreement** - Verbal definition/operationalization of the specific concept labels are entirely aligned in their content meaning, even if synonyms are used.
- **3: 75% Agreement** - Verbal definition/operationalization of the specific concept labels are largely aligned in content, with only minor differences.
- **2: 50% Agreement** - Verbal definition/operationalization of the specific concept labels show a moderate degree of alignment, with some significant differences.
- **1: 25% Agreement** - Verbal definition/operationalization of the specific concept labels show little alignment in content, with several notable differences.
- **0: No Agreement** - Verbal definition/operationalization of the specific concept labels show no content-based agreement.

Table 7.2: Evaluation of Rater Agreement on Definitions/Operationalizations of Concepts / Propositions / Operators (from Tables 3)

Please rate the agreement between the concepts/uncommon operators by entering the following number:
4 - Complete Agreement
3 - 75% Agreement
2 - 50% Agreement
1 - 25% Agreement
0 - No Agreement
NA - Not Applicable

Nr.	concept / proposition - Rater 1	concept / proposition - Rater 2	Rating Rater 1	Rating Rater 2	Comment
0	concept / proposition 1.1	concept / proposition 1.2	X	X	
	Verbal Definition 1.1	Verbal Definition 1.2			
	Operationalization 1.1	Operationalization 1.2			
1			X	X	

(7.3) Final Consensus Version

1) The two raters try to find a final consensus version of the Implications / Propositions they can agree on and list it in Table 7.3.1 below.
    a) Comments should be used to signal or communicate insights or remaining disagreement or uncertainty among the raters.
2) Based on Table 7.3.1 below the raters should now decide on a final version of the corresponding concept labels, verbal definitions, and operationalizations (Table 7.3.2). Table 7.3.1 determines the concept labels. The prior considerations documented in Table 7.2 should be taken into account during this process.
    a) Comments should be used to signal or communicate insights or remaining disagreement or uncertainty among the raters.
3) Auxiliary assumptions, Problems, Insights, Further directions and Open Questions can also be discussed and inserted in the tables (7.3.3-7.3.5) below.
4) **Full group discussion:** The same procedure can be followed when a final version is to be agreed upon by the larger group of all raters.

Table 7.3.1: Final Consensus Implications / Propositions

Nr	IF	Then	Reference / Comments / Markers
1			
2			

Table 7.3.2: Final Consensus Definitions/Operationalizations

Nr.	Concept / Proposition / Operator	Verbal Definition	Measurement/ Operationalization	Reference / Comments / Markers
1				
2				

Table 7.3.3: Final Consensus Auxiliary Assumptions (only if necessary - potentially skip)

Nr.	Description	Definition (Optional: Operationalization)	Reference / Comments / Markers
1			
2			
…			

Table 7.3.4: Final Consensus Problems

Nr.	Description	Reference / Comments / Markers
1		
2		
…		

Table 7.3.5: Final Consensus Insights, Further directions and Open Questions Table

Nr.	Description	Reference / Comments / Markers
1		
2		
…		

(8) Analysis of Theory Specification Issues and Classification of Implications

1) Please fill out the following table for the Final Consensus Implications / Propositions (Table 7.3.1), the Final Consensus Definitions/Operationalizations (Table 7.3.2) and the Final Consensus Problems (Table 7.3.4).

Table 8: Analysis of Theory Specification Issues and Classification of Implications

Please count and provide Answers in the format “x out of y”.

1) In how many implications did aspects have to be derived / imposed in the theory specification procedure since they were not specified by the theory? (You should have marked them with the marker “[DER]” (Table 7.3.1).

____ out of ___ implications derived / imposed

2) In how many definitions did aspects have to be derived / imposed / were indicated as missing in the theory specification procedure since they were not specified by the theory? (You should have marked them with the marker “[DER]” or [“MIS”] (Table 7.3.2).

____ out of ___ definitions derived / imposed / missing

3) In how many operationalizations did aspects have to be derived / imposed / were indicated as missing in the theory specification procedure since they were not specified by the theory? (You should have marked them with the marker “[DER]” or [“MIS”] (Table 7.3.2).

____ out of ___ operationalizations derived / imposed / missing

Provide counts for problems detected in Table 7.3.4:

4) the number of tautologies / tautology risks detected (marker: TAU): _____

5) the number of contradictions / contradictions risks detected (marker: CON): _____

6) the number of underspecifications / underspecification risks detected (marker: USPEC): _____